1. Introduction to Stored Procedures

A stored procedure is a named, precompiled collection of one or more SQL statements that is stored inside the database and can be executed (called) whenever required. Instead of sending a long block of SQL from an application every time, the application simply calls the procedure by name.

Stored procedures behave like functions or subroutines in a programming language — they can accept parameters, perform logic (loops, conditions, variable declarations), and return results.

Why use Stored Procedures?

- Performance: Precompiled and cached by the database engine, so repeated execution is faster than sending raw SQL each time.
- Reusability: Write the logic once, call it from many applications or scripts.
- Security: Users can be granted EXECUTE permission on a procedure without granting direct access to the underlying tables.
- Reduced network traffic: A single CALL statement replaces many lines of SQL sent over the network.
- Maintainability: Business logic is centralized in the database, so changes are made in one place.
- Consistency: Ensures the same validation/business rules are applied every time, regardless of which application calls it.

2. CREATE PROCEDURE Syntax

The general syntax (MySQL dialect) to create a stored procedure is:

```
DELIMITER //

CREATE PROCEDURE procedure_name (
    [IN | OUT | INOUT] parameter_name data_type, ...
)
BEGIN
    -- SQL statements / logic
    -- declarations, conditions, loops, etc.
```

END //

DELIMITER ;

Key points:

- DELIMITER // temporarily changes the statement terminator from ';' to '//' so that the semicolons inside the procedure body do not end the CREATE PROCEDURE statement prematurely.
- The procedure body is enclosed between BEGIN and END.
- Parameters are optional — a procedure can be created with zero, one, or many parameters.
- After the procedure is created, DELIMITER is reset back to ';'.

Advantages

Faster execution after compilation.

Reduced duplicate SQL.

Supports transactions, loops, conditions, variables, cursors, and exception handlers.

Easy maintenance.

Disadvantages

Debugging can be difficult.

Vendor-specific syntax.

Complex procedures may become hard to maintain.

3. IN, OUT, and INOUT Parameters

Parameter Type	Direction	Behaviour
IN	Caller -> Procedure	Default type. Passes a value INTO the procedure. Changes made inside do NOT reflect back to the caller.

OUT	Procedure -> Caller	Procedure sets a value that is returned OUT to the caller. Initial value passed in is ignored.
INOUT	Both directions	Caller passes a value IN, procedure can modify it, and the updated value is returned back OUT.

NoParameters

==========

1. Display All Employees

Question:

Write a stored procedure to display all employee records.

Answer:

```
DELIMITER //

CREATE PROCEDURE GetAllEmployees()
BEGIN
   SELECT * FROM Employees;
END //

DELIMITER ;
```

Call

```
CALL GetAllEmployees();
SHOW PROCEDURE STATUS WHERE Db = DATABASE();
```

---

2. Display IT Department Employees

Question:

Write a procedure to display employees working in the IT department.

Answer

```
DELIMITER //

CREATE PROCEDURE GetITEmployees()
BEGIN
   SELECT * FROM Employees
   WHERE department='IT';
END //

DELIMITER ;
Call
CALL GetITEmployees();
```

---

3. Display Employees with Salary Greater Than 60000

Question:

Write a procedure to display employees whose salary is greater than 60000.

Answer

```
DELIMITER //

CREATE PROCEDURE HighSalaryEmployees()
BEGIN
   SELECT * FROM Employees
   WHERE salary > 60000;
END //

DELIMITER ;
Call
CALL HighSalaryEmployees();
```

---

4. Display Employees from Hyderabad

Question:

Write a procedure to display employees from Hyderabad.

Answer

```
DELIMITER //

CREATE PROCEDURE HyderabadEmployees()
BEGIN
   SELECT * FROM Employees
   WHERE city='Hyderabad';
END //

DELIMITER ;
```

Call

```
CALL HyderabadEmployees();
```

---

5. Display Employees with Experience Greater Than 5 Years

Question:

Write a procedure to display employees having more than 5 years of experience.

Answer

```
DELIMITER //

CREATE PROCEDURE ExperiencedEmployees()
BEGIN
   SELECT * FROM Employees
   WHERE experience > 5;
END //
```

```
DELIMITER ;

Call

CALL ExperiencedEmployees();
```

---

6. Display Employee Names Only

Question:

Write a procedure to display only employee names.

Answer

```
DELIMITER //


CREATE PROCEDURE EmployeeNames()

BEGIN

   SELECT emp_name

   FROM Employees;

END //


DELIMITER ;

Call

CALL EmployeeNames();
```

---

7. Count Total Employees

Question:

Write a procedure to count the total number of employees.

Answer

```
DELIMITER //


CREATE PROCEDURE TotalEmployees()

BEGIN
```

```
    SELECT COUNT(*) AS TotalEmployees
    FROM Employees;
END //

DELIMITER ;
Call
CALL TotalEmployees();
```

---

8. Find Maximum Salary

Question:

Write a procedure to display the maximum salary.

Answer

```
DELIMITER //

CREATE PROCEDURE MaximumSalary()
BEGIN
    SELECT MAX(salary) AS HighestSalary
    FROM Employees;
END //

DELIMITER ;
Call
CALL MaximumSalary();
```

---

9. Find Minimum Salary

Question:

Write a procedure to display the minimum salary.

Answer

```
DELIMITER //

CREATE PROCEDURE MinimumSalary()
BEGIN
   SELECT MIN(salary) AS LowestSalary
   FROM Employees;
END //

DELIMITER ;
Call
CALL MinimumSalary();
```

---

10. Calculate Average Salary

Question:

Write a procedure to calculate the average salary.

Answer

```
DELIMITER //

CREATE PROCEDURE AverageSalary()
BEGIN
   SELECT AVG(salary) AS AverageSalary
   FROM Employees;
END //

DELIMITER ;
Call
CALL AverageSalary();
```

---

11. Display Employees Ordered by Salary

Question:

Write a procedure to display employees in ascending order of salary.

Answer

```
DELIMITER //

CREATE PROCEDURE EmployeesBySalary()
BEGIN
   SELECT *
   FROM Employees
   ORDER BY salary;
END //

DELIMITER ;
```

Call

```
CALL EmployeesBySalary();
```

---

12. Display Employees Ordered by Experience

Question:

Write a procedure to display employees sorted by experience in descending order.

Answer

```
DELIMITER //

CREATE PROCEDURE EmployeesByExperience()
BEGIN
   SELECT *
   FROM Employees
   ORDER BY experience DESC;
```

```
END //

DELIMITER ;
Call
CALL EmployeesByExperience();
```

---

13. Display Finance Department Employees

Question:

Write a procedure to display Finance department employees.

Answer

```
DELIMITER //

CREATE PROCEDURE FinanceEmployees()
BEGIN
   SELECT *
   FROM Employees
   WHERE department='Finance';
END //

DELIMITER ;
Cal
lCALL FinanceEmployees();
```

---

14. Display HR Employees

Question:

Write a procedure to display all HR employees.

Answer

```
DELIMITER //
```

```
CREATE PROCEDURE HREmployees()
BEGIN
   SELECT *
   FROM Employees
   WHERE department='HR';
END //

DELIMITER ;
Call
CALL HREmployees();
```

---

15. Total Salary Paid

Question:

Write a procedure to calculate the total salary paid to employees.

Answer

```
DELIMITER //

CREATE PROCEDURE TotalSalary()
BEGIN
   SELECT SUM(salary) AS TotalSalary
   FROM Employees;
END //

DELIMITER ;
Call
CALL TotalSalary();
```

---

16. Employees with Salary Between 50000 and 80000

Question:

Write a procedure to display employees whose salary is between 50000 and 80000.

Answer

```
DELIMITER //

CREATE PROCEDURE SalaryRange()
BEGIN
   SELECT *
   FROM Employees
   WHERE salary BETWEEN 50000 AND 80000;
END //

DELIMITER ;
Call
CALL SalaryRange();
```

---

17. Employees from Bangalore

Question:

Write a procedure to display employees from Bangalore.

Answer

```
DELIMITER //

CREATE PROCEDURE BangaloreEmployees()
BEGIN
   SELECT *
   FROM Employees
   WHERE city='Bangalore';
```

```
END //

DELIMITER ;
Call
CALL BangaloreEmployees();
```

---

18. Employees Having Salary Above Average

Question:

Write a procedure to display employees whose salary is greater than the average salary.

Answer

```
DELIMITER //

CREATE PROCEDURE AboveAverageSalary()
BEGIN
   SELECT *
   FROM Employees
   WHERE salary > (SELECT AVG(salary) FROM Employees);
END //

DELIMITER ;
Call
CALL AboveAverageSalary();
```

---

19. Department-wise Employee Count

Question:

Write a procedure to display the number of employees in each department.

Answer

```
DELIMITER //
```

```
CREATE PROCEDURE DepartmentWiseCount()
BEGIN
   SELECT department,
        COUNT(*) AS TotalEmployees
   FROM Employees
   GROUP BY department;
END //

DELIMITER ;
Call
CALL DepartmentWiseCount();
```

---

20. Employees with Maximum Experience

Question:

Write a procedure to display the employee(s) having the highest experience.

Answer

```
DELIMITER //

CREATE PROCEDURE HighestExperience()
BEGIN
   SELECT *
   FROM Employees
   WHERE experience = (
      SELECT MAX(experience)
      FROM Employees
   );
END //
```

DELIMITER ;

Call

CALL HighestExperience();

IN Parameter Procedures

Definition

An IN parameter is used to pass a value into a stored procedure.

The procedure can use the value, but changes made to the parameter inside the procedure are not returned to the caller.

Syntax

```
CREATE PROCEDURE procedure_name(
   IN parameter_name datatype
)
BEGIN
   SQL statements;
END;
```

Simple flow

Value

↓

IN Parameter

↓

Stored Procedure

1. Employee by ID

DELIMITER //

```
CREATE PROCEDURE GetEmployeeById(IN eid INT)
BEGIN
    SELECT * FROM Employees
    WHERE emp_id = eid;
END //

DELIMITER ;
CALL GetEmployeeById(101);
```

---

2. Employees by Department

```
DELIMITER //

CREATE PROCEDURE GetEmployeesByDepartment(IN dept VARCHAR(30))
BEGIN
    SELECT *
    FROM Employees
    WHERE department = dept;
END //

DELIMITER ;
CALL GetEmployeesByDepartment('IT');
```

---

3. Employees by Salary

```
DELIMITER //

CREATE PROCEDURE GetEmployeesBySalary(IN sal DECIMAL(10,2))
BEGIN
    SELECT *
```

```
    FROM Employees
    WHERE salary >= sal;
END //

DELIMITER ;
CALL GetEmployeesBySalary(60000);
```

---

4. Employees by City

```
DELIMITER //

CREATE PROCEDURE GetEmployeesByCity(IN cityName VARCHAR(30))
BEGIN
    SELECT *
    FROM Employees
    WHERE city = cityName;
END //

DELIMITER ;
CALL GetEmployeesByCity('Hyderabad');
```

---

5. Employees by Experience

```
DELIMITER //

CREATE PROCEDURE GetExperiencedEmployees(IN exp INT)
BEGIN
    SELECT *
    FROM Employees
    WHERE experience >= exp;
```

```
END //

DELIMITER ;
CALL GetExperiencedEmployees(5);
```

---

6. Employee by Name

```
DELIMITER //

CREATE PROCEDURE GetEmployeeByName(IN ename VARCHAR(50))
BEGIN
	SELECT *
	FROM Employees
	WHERE emp_name = ename;
END //

DELIMITER ;
CALL GetEmployeeByName('Rahul');
```

---

7. Salary Range

```
DELIMITER //

CREATE PROCEDURE SalaryRange(IN minSal DECIMAL(10,2), IN maxSal DECIMAL(10,2))
BEGIN
	SELECT *
	FROM Employees
	WHERE salary BETWEEN minSal AND maxSal;
END //
```

```
DELIMITER ;
CALL SalaryRange(40000,70000);
```

---

8. Department and Experience

```
DELIMITER //

CREATE PROCEDURE DeptExperience(IN dept VARCHAR(30), IN exp INT)
BEGIN
	SELECT *
	FROM Employees
	WHERE department=dept
	AND experience>=exp;
END //

DELIMITER ;
CALL DeptExperience('IT',5);
```

---

9. City and Salary

```
DELIMITER //

CREATE PROCEDURE CitySalary(IN cityName VARCHAR(30), IN sal DECIMAL(10,2))
BEGIN
	SELECT *
	FROM Employees
	WHERE city=cityName
	AND salary>=sal;
END //
```

```
DELIMITER ;
CALL CitySalary('Hyderabad',60000);
```

---

10. Finance Employees

```
DELIMITER //

CREATE PROCEDURE FinanceEmployees()
BEGIN
        SELECT *
        FROM Employees
        WHERE department='Finance';
END //

DELIMITER ;
CALL FinanceEmployees();
```

OUT Parameter Procedures

Definition

An OUT parameter is used to return a value from the stored procedure to the caller.

The caller provides a variable to receive the result.

```
CREATE PROCEDURE procedure_name(
   OUT parameter_name datatype
)
BEGIN
   SET parameter_name = value;
END;
```

Usually, we call an OUT parameter using a session variable:

SET @result = 0;

CALL procedure_name(@result);

SELECT @result;

1. Total Employees

```
DELIMITER //

CREATE PROCEDURE TotalEmployees(OUT total INT)
BEGIN
    SELECT COUNT(*) INTO total
    FROM Employees;
END //

DELIMITER ;
CALL TotalEmployees(@t);
SELECT @t;
```

---

2. Maximum Salary

```
DELIMITER //

CREATE PROCEDURE MaximumSalary(OUT maxSal DECIMAL(10,2))
BEGIN
    SELECT MAX(salary)
    INTO maxSal
    FROM Employees;
```

```
END //

DELIMITER ;
CALL MaximumSalary(@m);
SELECT @m;
```

---

3. Minimum Salary

```
DELIMITER //

CREATE PROCEDURE MinimumSalary(OUT minSal DECIMAL(10,2))
BEGIN
	SELECT MIN(salary)
	INTO minSal
	FROM Employees;
END //

DELIMITER ;
```

---

4. Average Salary

```
DELIMITER //

CREATE PROCEDURE AverageSalary(OUT avgSal DECIMAL(10,2))
BEGIN
	SELECT AVG(salary)
	INTO avgSal
	FROM Employees;
END //
```

```
DELIMITER ;
```

---

5. Total Salary

```
DELIMITER //

CREATE PROCEDURE TotalSalary(OUT totalSal DECIMAL(10,2))
BEGIN
    SELECT SUM(salary)
    INTO totalSal
    FROM Employees;
END //

DELIMITER ;
```

---

6. Highest Experience

```
DELIMITER //

CREATE PROCEDURE HighestExperience(OUT exp INT)
BEGIN
    SELECT MAX(experience)
    INTO exp
    FROM Employees;
END //

DELIMITER ;
```

---

7. Lowest Experience

```
DELIMITER //
```

```
CREATE PROCEDURE LowestExperience(OUT exp INT)
BEGIN
	SELECT MIN(experience)
	INTO exp
	FROM Employees;
END //

DELIMITER ;
```

---

8. IT Employee Count

```
DELIMITER //

CREATE PROCEDURE ITEmployeeCount(OUT cnt INT)
BEGIN
	SELECT COUNT(*)
	INTO cnt
	FROM Employees
	WHERE department='IT';
END //

DELIMITER ;
```

---

9. Hyderabad Employee Count

```
DELIMITER //

CREATE PROCEDURE HyderabadCount(OUT cnt INT)
BEGIN
```

```
    SELECT COUNT(*)
    INTO cnt
    FROM Employees
    WHERE city='Hyderabad';
END //

DELIMITER ;
```

---

10. Average Experience

```
DELIMITER //

CREATE PROCEDURE AverageExperience(OUT avgExp DECIMAL(5,2))
BEGIN
    SELECT AVG(experience)
    INTO avgExp
    FROM Employees;
END //

DELIMITER ;
```

INOUT Parameter Procedures

INOUT Parameter

Definition

An INOUT parameter is used when we want to send a value into the procedure and receive the modified value back.

In other words:

INOUT = Input + Output

Syntax

```
CREATE PROCEDURE procedure_name(
   INOUT parameter_name datatype
)
BEGIN
   SET parameter_name = new_value;
END;
```

Calling:

```
SET @value = initial_value;

CALL procedure_name(@value);

SELECT @value;
```

Flow

@value

↓

IN

↓

PROCEDURE

↓

modified value

↓

OUT

↓

@value

1. Add Bonus

```
DELIMITER //

CREATE PROCEDURE AddBonus(
        IN empid INT,
        INOUT bonus DECIMAL(10,2)
)
BEGIN
        UPDATE Employees
        SET salary = salary + bonus
        WHERE emp_id = empid;

        SELECT salary
        INTO bonus
        FROM Employees
        WHERE emp_id = empid;
END //

DELIMITER ;
SET @b=5000;
CALL AddBonus(101,@b);
SELECT @b;
```

---

2. Deduct Salary

```
DELIMITER //
```

```
CREATE PROCEDURE DeductSalary(
        IN empid INT,
        INOUT amount DECIMAL(10,2)
)
BEGIN
        UPDATE Employees
        SET salary=salary-amount
        WHERE emp_id=empid;

        SELECT salary
        INTO amount
        FROM Employees
        WHERE emp_id=empid;
END //

DELIMITER ;
SET @deduction = 5000;
CALL DeductSalary(101,@deduction);
SELECT @deduction;
```

---

3. Increase Experience

```
DELIMITER //

CREATE PROCEDURE IncreaseExperience(
        IN empid INT,
        INOUT exp INT
)
BEGIN
```

```
        UPDATE Employees
        SET experience=experience+exp
        WHERE emp_id=empid;

        SELECT experience
        INTO exp
        FROM Employees
        WHERE emp_id=empid;
END //

DELIMITER ;
SET @exp = 2;
CALL IncreaseExperience(101,@exp);
SELECT @exp;
```

---

5. Increase Salary by Amount

```
DELIMITER //

CREATE PROCEDURE SalaryIncrement(
        IN empid INT,
        INOUT inc DECIMAL(10,2)
)
BEGIN
        UPDATE Employees
        SET salary=salary+inc
        WHERE emp_id=empid;
```

```
        SELECT salary
        INTO inc
        FROM Employees
        WHERE emp_id=empid;
END //

DELIMITER ;
```

Stored Procedures - Scenario Based Questions (MySQL)

These questions are commonly asked in interviews and practical assessments. Each scenario requires creating a stored procedure.

Scenario 1: Employee Search

Problem

A company wants to search employee details by Employee ID without writing the same query repeatedly.

Task

Create a stored procedure that accepts Employee ID and displays employee details.

Concept: IN Parameter

Scenario 2: Department Employees

Problem

The HR department wants to see all employees working in a specific department.

Task

Create a stored procedure that accepts the department name and displays all employees.

Concept: IN Parameter

Scenario 3: Salary Filter

Problem

The finance team wants to identify employees earning more than a specified salary.

Task

Create a stored procedure that accepts a salary amount and displays all employees earning more than that amount.

Concept: IN Parameter

Scenario 4: City-wise Employees

Problem

The company has offices in multiple cities. Management wants to view employees from a particular city.

Task

Create a stored procedure that accepts a city name and displays employees from that city.

Concept: IN Parameter

Scenario 5: Experience-Based Employees

Problem

The recruitment team wants employees who have at least a certain number of years of experience.

Task

Create a stored procedure that accepts years of experience and displays matching employees.

Concept: IN Parameter

Scenario 6: Total Employees

Problem

The CEO wants to know the total number of employees in the company.

Task

Create a stored procedure that returns the total employee count.

Concept: OUT Parameter

Scenario 7: Highest Salary

Problem

The finance department wants to know the highest salary paid in the company.

Task

Create a stored procedure that returns the highest salary.

Concept: OUT Parameter

Scenario 8: Average Salary

Problem

The accounts department wants to calculate the average salary of all employees.

Task

Create a stored procedure that returns the average salary.

Concept: OUT Parameter

Scenario 9: Total Salary Expenditure

Problem

Management wants to know the total monthly salary expenditure.

Task

Create a stored procedure that returns the sum of all employee salaries.

Concept: OUT Parameter

Scenario 10: IT Employee Count

Problem

The IT manager wants to know how many employees are working in the IT department.

Task

Create a stored procedure that returns the number of IT employees.

Concept: OUT Parameter

Scenario 11: Bonus Distribution

Problem

The company announces a ₹5,000 bonus for a specific employee.

Task

Create a stored procedure that accepts Employee ID and Bonus Amount, updates the salary, and returns the updated salary.

Concept: INOUT Parameter

Scenario 12: Salary Increment

Problem

Every year, employees receive an increment.

Task

Create a stored procedure that accepts Employee ID and Increment Amount, updates the salary, and returns the new salary.

Concept: INOUT Parameter

Scenario 13: Tax Deduction

Problem

The payroll department deducts tax from employees' salaries.

Task

Create a stored procedure that accepts Employee ID and Tax Amount, deducts the tax, and returns the updated salary.

Concept: INOUT Parameter

Scenario 14: Promotion

Problem

An employee gets promoted, resulting in both a salary increase and an increase in experience.

Task

Create a stored procedure that updates the employee's salary and experience and returns the updated salary.

Concept: INOUT Parameter

Scenario 15: Festival Bonus

Problem

During a festival, employees receive an additional bonus.

Task

Create a stored procedure that adds the bonus to the salary and returns the revised salary.

Concept: INOUT Parameter

Scenario 16: Student Results

Problem

A college wants to display all students who scored above a specified mark.

Task

Create a stored procedure that accepts the minimum marks and displays qualifying students.

Concept: IN Parameter

Scenario 17: Product Search

Problem

An e-commerce website allows users to search products by category.

Task

Create a stored procedure that accepts the category name and displays matching products.

Concept: IN Parameter

Scenario 18: Order Summary

Problem

An online shopping system wants to know the total number of orders placed.

Task

Create a stored procedure that returns the total number of orders.

Concept: OUT Parameter

Scenario 19: Bank Account Deposit

Problem

A customer deposits money into a bank account.

Task

Create a stored procedure that accepts Account Number and Deposit Amount, updates the balance, and returns the updated balance.

Concept: INOUT Parameter

Scenario 20: Hospital Patient Search

Problem

A hospital receptionist wants to retrieve patient details using the Patient ID.

Task

Create a stored procedure that accepts Patient ID and displays the patient's information.

Concept: IN Parameter

Scenario 21: Library Management

Problem

A librarian wants to know how many books are currently available.

Task

Create a stored procedure that returns the total number of available books.

Concept: OUT Parameter

Scenario 22: Hotel Room Booking

Problem

A hotel receptionist books a room for a customer.

Task

Create a stored procedure that updates the room status from **Available** to **Booked** and returns the updated status.

Concept: INOUT Parameter

Scenario 23: Inventory Management

Problem

A store sells a product, reducing the available stock.

Task

Create a stored procedure that accepts Product ID and Quantity Sold, updates the stock, and returns the remaining stock.

Concept: INOUT Parameter

Scenario 24: Employee Transfer

Problem

An employee is transferred to another department.

Task

Create a stored procedure that accepts Employee ID and New Department, updates the department, and displays the updated employee details.

Concept: IN Parameter

Scenario 25: Payroll System

Problem

The payroll department wants to identify employees whose salary falls within a specified range.

Task

Create a stored procedure that accepts a minimum salary and maximum salary and displays all matching employees.

Concept: IN Parameter

Summary

Scenario	Concept
Employee Search	IN
Department Employees	IN
Salary Filter	IN
City-wise Employees	IN
Experience Search	IN
Total Employees	OUT
Highest Salary	OUT
Average Salary	OUT
Total Salary	OUT
IT Employee Count	OUT
Bonus Distribution	INOUT
Salary Increment	INOUT

Tax Deduction	INOUT
Promotion	INOUT
Festival Bonus	INOUT
Student Results	IN
Product Search	IN
Order Summary	OUT
Bank Deposit	INOUT
Hospital Search	IN
Library Books Count	OUT
Hotel Booking	INOUT
Inventory Update	INOUT
Employee Transfer	IN
Payroll Salary Range	IN